

The Big Nine: How the Tech Titans and Their Thinking Machines Could Warp Humanity

Amy Webb (2019)

Stelle: 77

The lack of nuance is one part of AI's genesis problem: some dramatically overestimate the applicability of AI, while others argue it will become an unstoppable weapon.

Stelle: 95

In the United States, Google, Amazon, Apple, IBM, Microsoft and Facebook are hamstrung by the relentless short-term demands of capitalistic society, while in China, Baidu, Alibaba and Tencent are fueling the government's authoritarian ambitions.

Stelle: 107

So in place of coordinated national strategies to build organizational capacity inside the government, to build and strengthen our international alliances, and to prepare our military for the future of warfare, the United States has subjugated AI to the revolving door of politics. Instead of funding basic research into AI, the federal government has effectively outsourced R&D to the commercial sector and the whims of Wall Street.

Stelle: 113

individual city governments, well-capitalized advisory groups, and presidential candidates for the 2020 election have published their own punitive and, arguably, unenforceable guidelines on recommended AI ethics, values and rules. In every case, these groups are imposing their narrow ideas and worldviews on the big tech giants without asking for their input.

Stelle: 125

In July 2017, the Chinese government unveiled its Next Generation Artificial Intelligence Development Plan to become the global leader in AI by the year 2030 with a domestic industry worth at least \$150 billion,¹ which involved devoting part of its sovereign wealth fund to new labs and startups, as well as new schools launching specifically to train China's next generation of AI talent.

Stelle: 154

An AI-powered Social Credit System, according to a slogan in official planning documents, was developed to engineer a problem-free society by "allow(ing) the trustworthy to roam everywhere under heaven while making it hard for the discredited to take a single step."⁵ To promote "trustworthiness," citizens are rated on a number of different data points, like heroic acts (points earned) or traffic tickets (points deducted). Those with lower scores face hurdles applying for jobs, buying a home, or getting kids into schools. In some cities, high-scoring residents have their pictures on display.⁶ In other cities, such as Shandong, citizens who jaywalk have their faces publicly shared on digital billboards and sent automatically to Weibo, a popular social network.

Stelle: 185

As the structures and systems that govern society come to rely on AI, we will find that decisions being made on our behalf make perfect sense to machines—just not to us.

Stelle: 187

We've started to pass some major milestones in the technical and geopolitical development of AI, yet with every new advancement, AI becomes more invisible to us. The ways in which our data is being mined and refined is less obvious, while our ability to understand how autonomous systems make decisions grows less transparent.

Stelle: 189

This is particularly bad because at the moment there is no singular entity that can be held accountable for AI's development.

Stelle: 195

We have, therefore, a chasm in understanding of how AI is impacting daily life in the present, one growing exponentially as we move years and decades into the future. Shrinking that distance as much as possible through a critique of the developmental track that AI is currently on is my mission for this book.

Stelle: 200

What happens to society when we transfer power to a system built by a small group of people that is designed to make decisions for everyone? What happens when those decisions are biased toward market forces or an ambitious political party? The answer is reflected in the future opportunities we have, the ways in which we are denied access, the social conventions within our societies, the rules by which our economies operate, and even the way we relate to other people.

Stelle: 741

By not using any human data or expertise, Zero's creators removed the constraints of human knowledge on artificial intelligence. Humans, as it turned out, would have held the system back. The achievement was architecting a system that had the ability to think in an entirely new way and to make its own choices.⁴¹ It was a sudden, unexpected leap, one that portended a future in which AI systems could look at cancer screenings, evaluate climate data, and analyze poverty in nonhuman ways

Stelle: 779

Overwhelmingly, these teams who are architecting systems intended to make both choices and decisions are led by men. It's only a slightly more diverse group than the researchers who met at Dartmouth because of one big development: China. In recent years, China has become an important hub for AI, and that's because of a massive, government-funded effort at Chinese universities and at Baidu, Alibaba, and Tencent.

Stelle: 802

Thinking machines have creators, too—they are the new gods of AI, and they are mostly male, predominantly live in America, Western Europe, and China, and are tied, in some way, to the Big Nine. The soul of AI is a manifestation of their vision and intent for the future.

Stelle: 812

The future of AI is being built by a relatively few like-minded people within small, insulated groups. Again, I believe that these people are well intentioned. But as with all insulated groups that work closely together, their unconscious biases and myopia tend to become new systems of belief and accepted behaviors over time. What might have in the past felt unusual—wrong, even—becomes normalized as everyday thinking. And that thinking is what's being programmed into our machines.

Stelle: 830

AI's tribe has a familiar, catchy rallying cry: fail fast and fail often. In fact, a version of it — “move fast and break things” — was Facebook's official company motto until recently.

Stelle: 950

For a time, one of the more popular neural nets at universities was called Word2vec, and it was built by the Google Brain team. It was a two-layer system that processed text, turning words into numbers that AI could understand.¹⁷ For example, it learned that “man is to king as woman is to queen.” But the database also decided that “father is to doctor as mother is to nurse” and “man is to computer programmer as woman is to homemaker.”¹⁸ The very system students were exposed to was itself biased. If someone wanted to analyze the farther-reaching implications of sexist code, there weren't any classes where that learning could take place.

Stelle: 975

the Big Nine uses AI-powered software to sift through resumes, and it's trained to look for specific keywords describing hard skills. A portfolio of coursework outside the standard subjects would either be an anomaly or would render the applicant as invisible.

Stelle: 998

In universities, PhD candidates serve three functions: to collaborate on research, to teach undergraduate students, and to lead future work in their fields. Women receive only 23% of PhDs awarded in computer science, and only 28% awarded in mathematics and statistics, according to recent data from the National Center for Education Statistics.

Stelle: 1,001

So it should not come as a surprise that women received only 18% of undergraduate computer science degrees in recent years—and that's actually down from 37% in 1985.²² Black and Hispanic PhD candidates are woefully underrepresented—just 3% and 1% respectively.²³

Stelle: 1,243

While in the US, our tech giants are navigating a tricky path between national security and full transparency, the relationships the BAT has with China's government are exactly opposite. But here's a chilling example: the current posture of the US military is that a human must be kept in the loop, regardless of how advanced AI, unmanned systems, and robots become. This will ensure that we don't someday cede lethal authority to software. That is not the case in China.⁵⁶ PLA Lieutenant General Liu Guozhi, who directs the Chinese military's Science and Technology Commission, is quoted as warning, “(We) must... seize the opportunity to change paradigms.”

Stelle: 1,275

AI-powered directional microphones and smart cameras now dot the highways and streets of Shanghai. Drivers who honk excessively are automatically issued a ticket via Tencent's WeChat, while their names, photographs, and national identity card numbers are displayed on nearby LED billboards. If a driver pulls over on the side of the road for more than seven minutes, they will trigger another instant traffic ticket.⁶² It isn't just the ticket and the fine—points are deducted in the driver's social credit score. When enough points are deducted, they will find it hard to book airline tickets or land a new job.

Stelle: 1,299

Xi rejects the notions of market economies, a free internet, and a diverse ecosystem of competing and complementary ideas. China's tightly controlled domestic economy walls itself off from competition.

Stelle: 1,544

The Big Nine aren't just building hardware and code. They are building thinking machines that reflect humanity's values. The gap that currently exists between AI's tribes and everyday people is already causing worrying outcomes.

Stelle: 1,609

It should give us all pause that the Big Nine are building systems that fundamentally rely on people, and values articulating our aspirations for the improved quality of human life are not explicitly codified. If technological, economic, and social values aren't part of a company's statement of values, it is unlikely that the best interests of all of humanity will be prioritized during the research, design, and deployment process.

Stelle: 1,663

Conway's law, which says that in absence of stated rules and instructions, the choices teams make tend to reflect the implicit values of their tribe.

Stelle: 1,740

That inability to observe how AI is optimizing and making its decisions is what's known as the "black box problem." Right now, AI systems built by the Big Nine might offer open-source code, but they all function like proprietary black boxes.

Stelle: 1,776

How are humanity's billions of nuanced differences in culture, politics, religion, sexuality, and morality being optimized? In the absence of codified humanistic values, what happens when AI is optimized for someone who isn't anything like you?

Stelle: 1,921

Through its mining and refining of our data, the systems and techniques used to train machine-learning algorithms, and the optimization effect, the Big Nine are nudging at a grand scale. Even if it feels as if you have the ability to make a choice, what you're experiencing is an illusion. Nudging not only changes our relationship to technology—it is morphing our values in nearly imperceptible ways.

Stelle: 1,928

Optimizing AI means nudging humans. In other professional and technical fields, there is a set of guiding principles that governs how people work, and nudging tends to violate the spirit of those principles. In medicine, there is the Hippocratic oath, which requires physicians to swear to uphold specific ethical standards. Lawyers adhere to attorney-client privilege and to confidentiality, which protect the conversations people have with the professionals who are representing them. Journalists abide by many guiding principles, which include standards like using primary source information and reporting on stories in the public interest. Right now, no one is incentivized to consider the unforeseen costs of optimizing AI in the absence of codified, humanistic principles.

Stelle: 1,971

Since AI isn't being taught to make perfect decisions, but rather to optimize, our response to changing forces in society matter a lot. Our values are not immutable. This is what makes the problem of AI's values so vexing. Building AI means predicting the values of the future. Our values aren't static. So how do we teach machines to reflect our values without influencing them?

Stelle: 1,980

The rules—the algorithm—by which every culture, society, and nation lives, and has ever lived, were always created by just a few people. Democracy, communism, socialism, religion, veganism, nativism, colonialism—these are constructs we've developed throughout history to help guide our decisions. Even in the best cases, they aren't future-proof.

Stelle: 1,988

There would be no way to create a set of commandments for AI. We couldn't write out all of the rules to correctly optimize for humanity, and that's because while thinking machines may be fast and powerful, they lack flexibility. There isn't an easy way to simulate exceptions, or to try and think through every single contingency in advance. Whatever rules might get written, there would always be a circumstance in the future in which some people might want to interpret the rules differently, or to ignore them completely, or to create amendments in order to manage an unforeseen circumstance.

Stelle: 2,020

In August 2019, IBM introduced an open source "AI Explainability" toolkit with algorithms for case-based reasoning, global explanations, and tools to address gender and racial bias. Noting that real people who use AI systems might not understand how training models work, the project was launched to make machine-based decision making more transparent.

Stelle: 2,025

The Big Nine produces ethics studies and white papers, it convenes experts to discuss ethics, and it hosts panels about ethics—but that effort is not intertwined enough with the day-to-day operations of the various teams working on AI.

Stelle: 2,034

There isn't a facet of your personal or professional life that won't be impacted by AI. What if, in a rush to get products to market or to please certain government officials, your values aren't reflected not just in AI but in all of the systems it touches? How comfortable are you now knowing that the BAT and G-MAFIA are making decisions affecting all of our futures?

Stelle: 2,070

Mapping out the potential impacts of AI now gives us agency in determining where human society goes from here: we can choose to maximize the good and minimize harm, but we cannot do this in reverse. Most often we do our critical thinking after a crisis as we try to reverse-engineer poor decisions, figure out how warning signs were missed, and find people and institutions to blame.

Stelle: 2,081

With regards to AI, there are now clear warning signs portending future crises, even if those signals are not immediately obvious.

Stelle: 2,082

Warning #1: We mistakenly treat artificial intelligence like a digital platform—similar to the internet—with no guiding principles or long-term plans for its growth. We have failed to recognize that AI has become a public good. When economists talk about a “public good,” they use a very strict definition: it must be nonexcludable, meaning it’s impossible to exclude someone from using it because to do so would be impossible, and it must be nonrivalrous, meaning that when one person uses it, another can use it too. Government services, like national defense, fire service, and trash pickup, are public goods. But public goods can also be created in markets, and as time wears on, market-borne public goods can produce unintended consequences.

Stelle: 2,106

Treating AI as a public good does not preclude the G-MAFIA from earning revenue and growing. It just means shifting our thinking and expectations. Someday, we will not have the luxury of debating and discussing automation within the context of human rights and geopolitics. AI will have become too complex for us to untangle and shape into something we prefer.

Stelle: 2,109

Warning #2: AI is rapidly concentrating power among the few, even as we view AI as an open ecosystem with few barriers.

Stelle: 2,129

we have asked and allowed the G-MAFIA to make serious and significant decisions that impact the future of our workforce, our national security, our economic growth, and our individual opportunities.

Stelle: 2,134

It’s difficult to wrap our heads around potential crises and opportunities before they’ve already happened, and that’s why we tend to stick to our existing narratives. That’s why we reference killer robots rather than paper cuts. Why we fetishize the future of AI rather than fearing the many algorithms that learn from our data.

Stelle: 2,165

Scenarios are a tool to help us cope with a cognitive bias behavioral economics and legal scholar Cass Sunstein calls “probability neglect.”⁹ Our human brains are bad at assessing risk and peril. We assume that common activities are safer than novel or uncommon activities.

Stelle: 2,264

if ASI’s cognitive abilities are orders of magnitude better than ours (remember, we’re just a few clicks above chimpanzees), then it would be impossible for us to imagine the consequences such powerful machines might have on our civilization.

Stelle: 2,336

The average person’s email address has been repurposed as a login; their mobile phone number is used to authenticate transactions; and their smartphone is used to locate them in the physical world. If you are a Gmail user, Google—and by extension its AIs—knows you better than your spouse or partner. It knows the names and email addresses of everyone you talk to, along with their demographic information (e.g., age, gender, location). Google knows when you tend to open email and under what circumstances. From your email, it knows your travel itineraries, your financial records, and what you buy. If you take photos with your Android phone, it knows the faces of your friends and family members, and it can detect anomalies to make inferences: for example, sudden new pics of the same person might indicate a new girlfriend (or an affair). It knows all of your meetings, doctor appointments, and plans to hit the gym. It knows whether you observe Ramadan or Rosh Hashanah, whether you’re a churchgoer, or whether you practice no religion at all. It knows where you should be on a given Tuesday afternoon, even if you’re somewhere else. It knows what you search for, using your fingers and your voice, and so it knows whether you’re miscarrying for the first time, learning how to make paella, struggling with your sexual identity or gender assignment, considering giving up meat, or looking for a new job. It cross-links all this data, learning from it and productizing and monetizing it as it nudges you in predetermined directions. Right now, Google knows all of this information because you’ve voluntarily linked it all to just one record—your Gmail address—which, by the way, you’ve probably also used to buy stuff on Amazon and to log into Facebook. This isn’t a complaint; it’s a fact of modern life. As AI advances, a more robust personal data record will afford greater efficiencies to the Big Nine, and so they will nudge us to accept and adopt PDRs, even if we don’t entirely understand the implications of using them. Of course, in China, PDRs are already being piloted under the auspices of its social credit score.

Stelle: 2,698

LEARNING TO LIVE WITH MILLIONS OF PAPER CUTS: THE PRAGMATIC SCENARIO By 2023, we've acknowledged AI's problems but along the way decided to make only minor tweaks in the developmental track of artificial intelligence, a system that we can all see is clearly fractured. We pursue only tweaks because AI's stakeholders aren't willing to get uncomfortable: to sacrifice financial gains, make politically unpopular choices, and curb our wild expectations in the short-term, even if it means improving our long-term odds of living alongside AI. Worse, we ignore China and its plans for the future. Leaders in Congress, our various federal agencies, and the White House continue to deprioritize artificial intelligence and advanced scientific research in general, preferring to invest in industries that are politically appealing but nearing obsolescence.

Stelle: 2,711

The Big Nine are forced to prioritize speed over safety, so AI's developmental track—from ANI to AGI and beyond—pushes ahead without first resolving serious technical vulnerabilities.

Stelle: 2,762

Another paper cut: malicious actors can inject poisonous data into AI's training programs. Neural networks are vulnerable to "adversarial examples," which are fake or intentionally designed with wrong information to cause an AI system to make a mistake.

Stelle: 2,772

A team within Google's Brain division published a paper in 2018 on how a bad actor could inject adversarial information into a computer vision database and effectively reprogram all the AI systems that learn from it.

Stelle: 2,776

A different Google Brain team discovered that adversarial information could also be used to generate new information that can be put to good use in what's called a "generative adversarial network," or GAN. In essence, it's the Turing test but without any humans involved.

Stelle: 2,817

Your personal data record is built, maintained, and owned initially by four of the G-MAFIA: Google, Amazon, Apple, and Facebook. But here's the rub: you're not even aware that PDRs exist or that they're being used by the G-MAFIA and by AI's tribes. It's not intentional but rather an oversight due to speed. It's all explained in the terms of service we all agree to but never, ever read. The formatting used by each PDR provider isn't complementary, so there's both duplicative data being spread around and, paradoxically, big holes with important data missing.

Stelle: 2,844

To avoid antitrust lawsuits, we're told that at any time, we can port our PDRs between operating systems. Of course, in practical terms, it's nearly impossible to make the change. You're reminded of trying to switch between iOS and Android many years ago when you discovered that lots of important data and settings were lost forever, progress within apps was erased, many apps didn't even work (and you could no longer get a refund), and all of the places you previously hosted your photos and videos couldn't be accessed easily. Now that your PDR is being used by third parties—such as schools, hospitals, and airlines—it's a far more difficult process to move between Google and Apple.

Stelle: 3,043

You, like all Americans, are learning to live with constant, low-grade anxiety. Our national sense of uneasiness is often compared to nuclear war threats in the 1960s and 1980s. Except that this time around, we're not sure what to fear, exactly. You don't know if your PDR is safeguarded or what personal data China might have access to. You're unsure of how deeply rooted Chinese government hackers are within our country's infrastructure systems.

Stelle: 3,110

"This is the way the world ends, not with a bang but a whimper." —T. S. ELIOT

Stelle: 3,112

By 2023, we have closed our eyes to artificial intelligence's developmental track. We missed all the signals, we ignored the warning signs, and we failed to actively plan for the future. We helped the Big Nine compete against itself as we indulged our consumerist desires, buying the latest gadgets and devices, celebrating every new opportunity to record our voices and faces, and submitting to an open pipeline that continually siphoned off our data.

Stelle: 3,131

Surveillance expands while simultaneously becoming less obvious. The line between our personal data and the data we generate at work blurs, and so do the criteria for who gets to use our data and when.

Stelle: 3,168

Chinese citizens are learning to live with automated monitoring and consequences of stepping out of line. Crime is down, and social unrest is curtailed, and for a time the middle and upper classes preserve the status quo. They have access to luxury clothing and handbags, designer furniture and statement cars never imagined by their parents and grandparents. Promises are made to lift all Chinese people out of poverty. For now at least, it seems like privacy, religious freedom, sexual identity, and free speech are reasonable trade-offs for earning a desirable social credit score.

Stelle: 3,279

The lack of interoperability between AI frameworks and systems led to segregation by PDR and household, and that is why we now have a digital caste system. By choosing Google, Apple, or Amazon, you are forced to align your family values with the values of the corporation. Apple families are rich, maybe a little less AI-savvy, and live in fancy houses. Google families might be rich and techy, or middle class and fine with marketing, or complacent enough that having a lot of choices in life doesn't matter all that much. There is no way to sugarcoat Amazon families: they're poor, even if they have free access to cool gadgets.

Stelle: 3,310

In America, society is beginning to feel uncomfortably Huxleian, as we acquiesce, get married, and have babies with our fellow Apples, or Google Blues, or Amazons.

Stelle: 3,347

AI bestows immense economic power on Google, Apple, and Amazon—and unimaginable geopolitical and military power on China. By the end of the 2030s, we realize that AI has developed along parallel trajectories, supporting capitalism in the West and China's brand of communism throughout Asia, Africa, and Latin America. America and its allies, who once celebrated in the G-MAFIA's successes, are living under a system of AI totalitarianism. Citizens throughout China and all the countries supported by China's direct investment and infrastructure find that they, too, are living under a pervasive apparatus of AI-powered punishment and reward.

Stelle: 3,357

From the very beginning, AI's tribes in the United States said they wanted us to live our best selves, to pursue creative endeavors, and to collaborate on humanity's biggest challenges. It was a utopian ideal born in the bubble of Silicon Valley, whose progenitors had completely lost touch with the outside world. All of these systems were built to make our lives easier but have instead emboldened our laziness. They've eroded our sense of productivity and purpose. We rely on systems to make decisions for us. We resign ourselves to limited choices. We are going through the preprogrammed motions of daily living, optimized by AGI for everyone on the planet.

Stelle: 3,491

The best way to engineer systematic change is to treat AI as a public good and to see to the creation of a new international entity. This body would be responsible for setting guidelines for AI by enforcing standards, testing advanced systems before their commercial release, and monitoring activity as AI progresses from narrow to general to super intelligence.

Stelle: 3,604

In order to drive innovation and advancement at scale, we must build internal capacity for research, testing, and deployment—and we need cohesion across departments. At the moment, AI is outsourced to government contractors and consultancies.

Stelle: 3,617

For perspective, in 2017 alone the G-MAFIA spent a combined \$63 billion on R&D—nearly five times the US government's total science and tech research budget.

Stelle: 3,618

But it also points to a bigger, thornier problem: if our government can't or won't fund basic research, then the G-MAFIA is stuck answering to Wall Street.

Stelle: 3,671

The US government has no blueprint articulating long-term research and development funding targets in priority sectors. The challenge is particularly acute at a critical time in the developmental cycle of fundamental science and technologies like AI with important societal implications that reach beyond the existing mandate of federal agencies. With regard to artificial intelligence in particular, there are numerous initiatives, cells and centers now working independently on the future of AI on behalf of the United States. However there is neither inter-agency collaboration on these efforts nor any coordinated effort to streamline goals, outcomes, R&D efforts and funding.

Stelle: 3,684

We need a new, centralized office championing strategic foresight.

Stelle: 3,718

it's really hard to make the case for a talented computer scientist to join the government or military, given what the G-MAFIA offer.

Stelle: 3,765

There is no standard for transparency either. In the United States, the G-MAFIA, along with the American Civil Liberties Union, the New America Foundation, and the Berkman Klein Center at Harvard are part of the Partnership on AI, which is meant to promote transparency in AI research. The partnership published a terrific set of recommendations to help guide AI research in a positive direction, but those tenets are not enforceable in any way—and they're not observed within all of the business units of the G-MAFIA. They're not observed within the BAT, either.

Stelle: 3,769

The Big Nine are using flawed corpora (training data sets) that are riddled with bias. This is public knowledge. The challenge is that improving the data and learning models is a big financial liability.

Stelle: 3,872

Universities should promote dual degrees in computer science and political science, philosophy, anthropology, international relations, political science, creative arts, theology, and sociology.

Stelle: 3,875

Schools must incentivize even tenured professors to include discussions of philosophy, bias, risk, and ethics in their courses, while accreditation agencies should incentivize and reward schools that can demonstrate a curriculum that puts ethics at the heart of computer science teaching.